

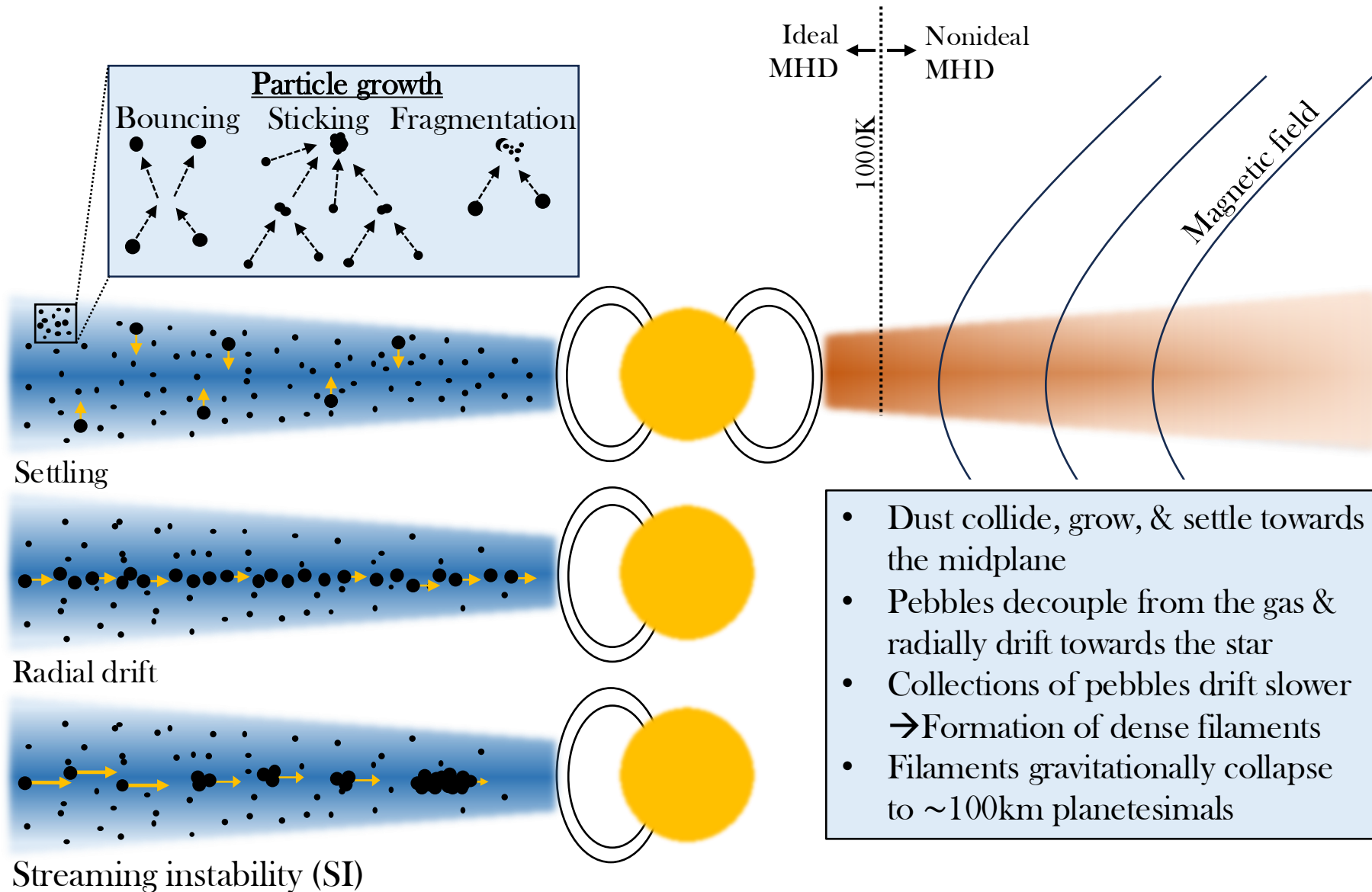
Planetesimal formation via the streaming instability persists under self-consistent MRI turbulence

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In collaboration with: Ziyang Xu, Jeonghoon Lim, Chao-Chin Yang, Pinghui Huang,
Mordecai-Mark Mac Low

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From dust to planetesimals



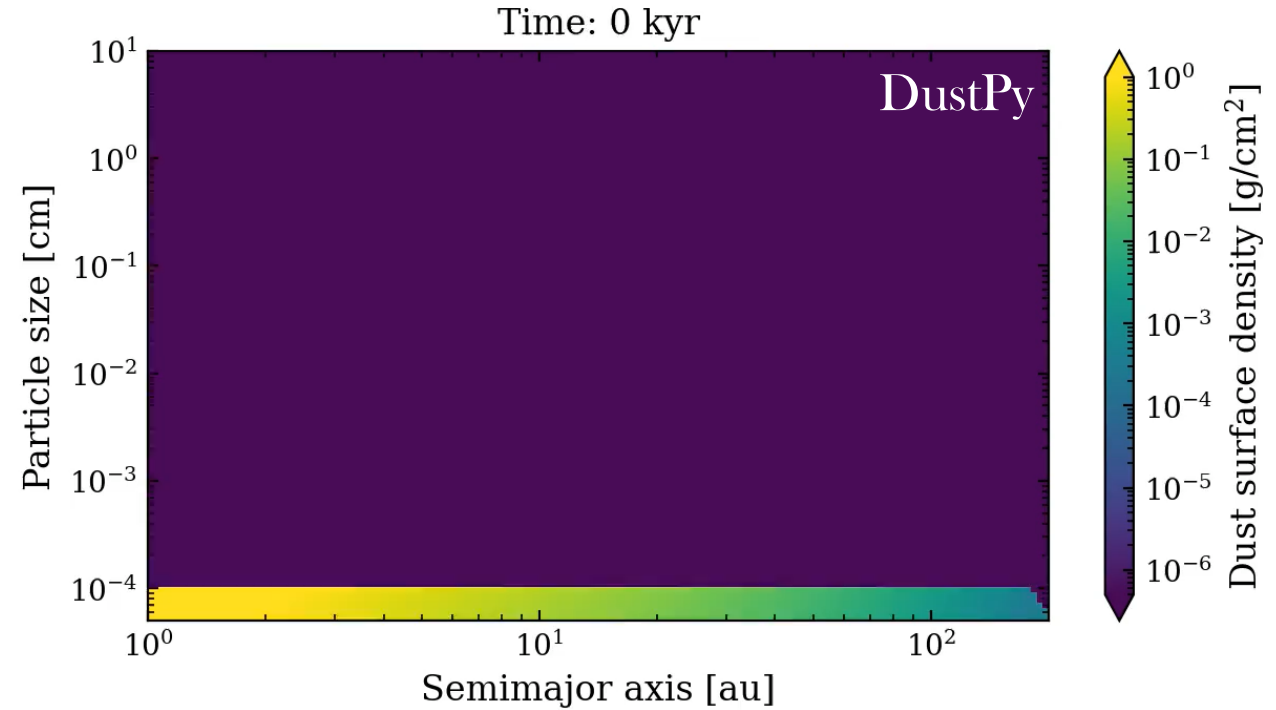
Modeling dust evolution

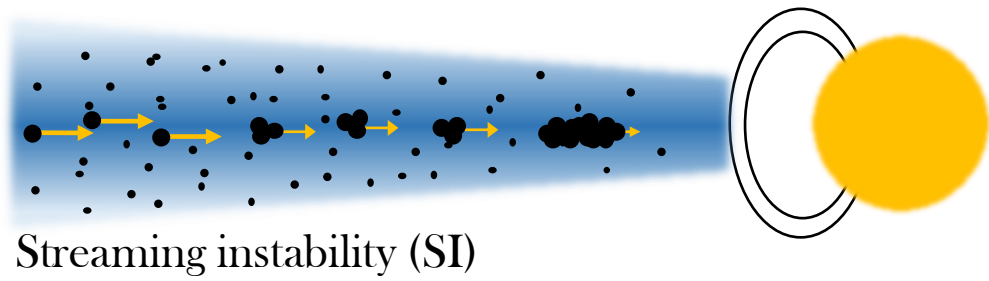
Dust treatment:

- Representative particles - Monte Carlo method
mcdust
- Particle size distribution - Smoluchowski coagulation eq.
DustPy, cuDisc, SHARK
- Semi-analytic models
pebble-predictor, two-pop-py, TriPoD, Ishinisan
- Coagulation in SPH codes – local
Phantom

Dust evolution comparison study (Eriksson et al. 2026):

- 1D radial: *two-pop-py, TriPoDPy, DustPy*
- 2D radial-vertical: *TriPoD (athena++), cuDisc, mcdust*





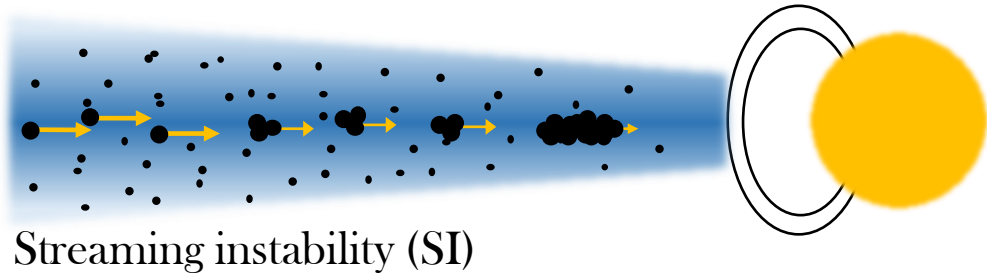
clumping of dust



Simulating the formation of planetesimals

Modeling the streaming instability requires 2D or 3D simulations with high spatial resolution
→ Global SI simulations across disk lifetimes are unfeasible

Derive planetesimal formation criteria from local models
→ Use these criteria in global models

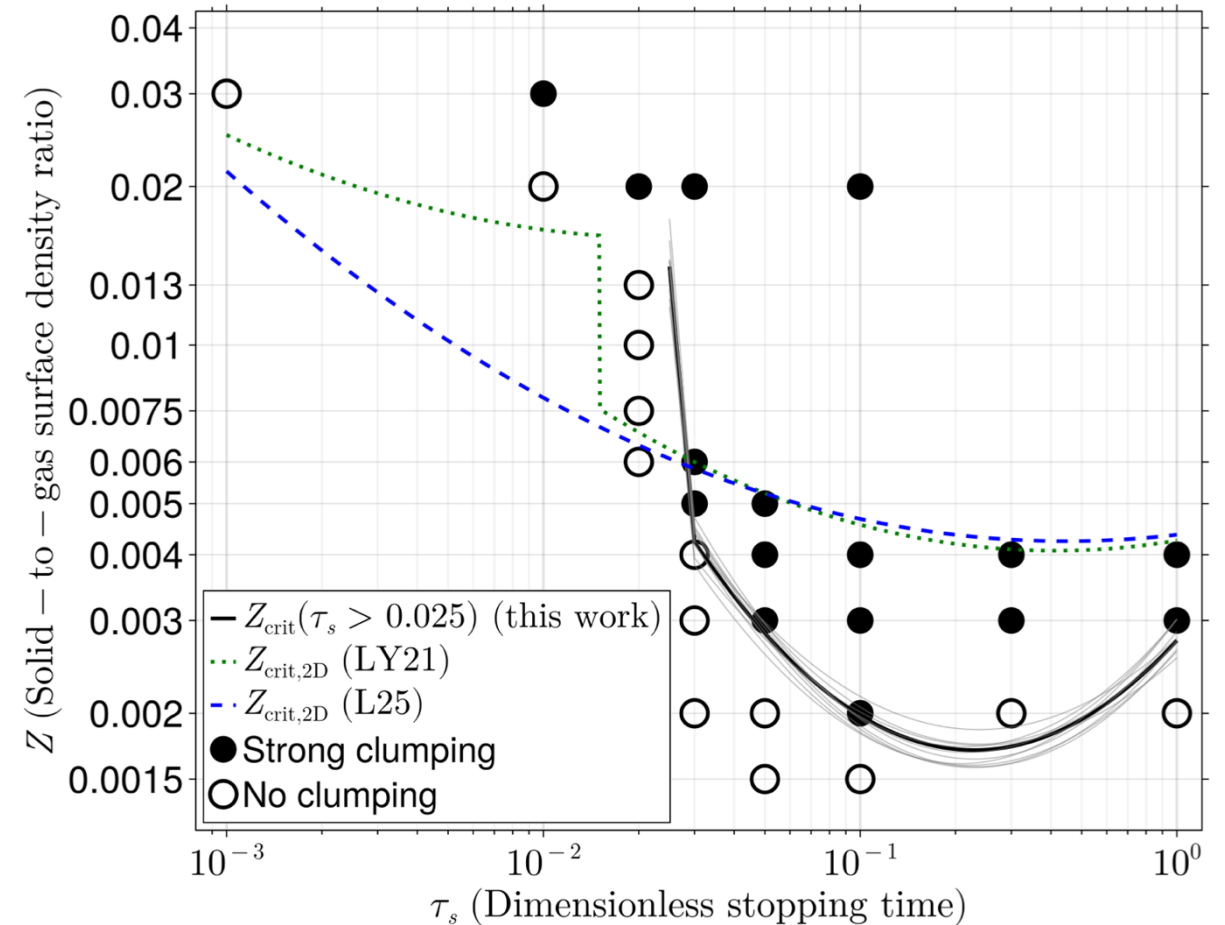


No external turbulence
Constant pressure gradient
Single particle sizes

Overview of clump criteria

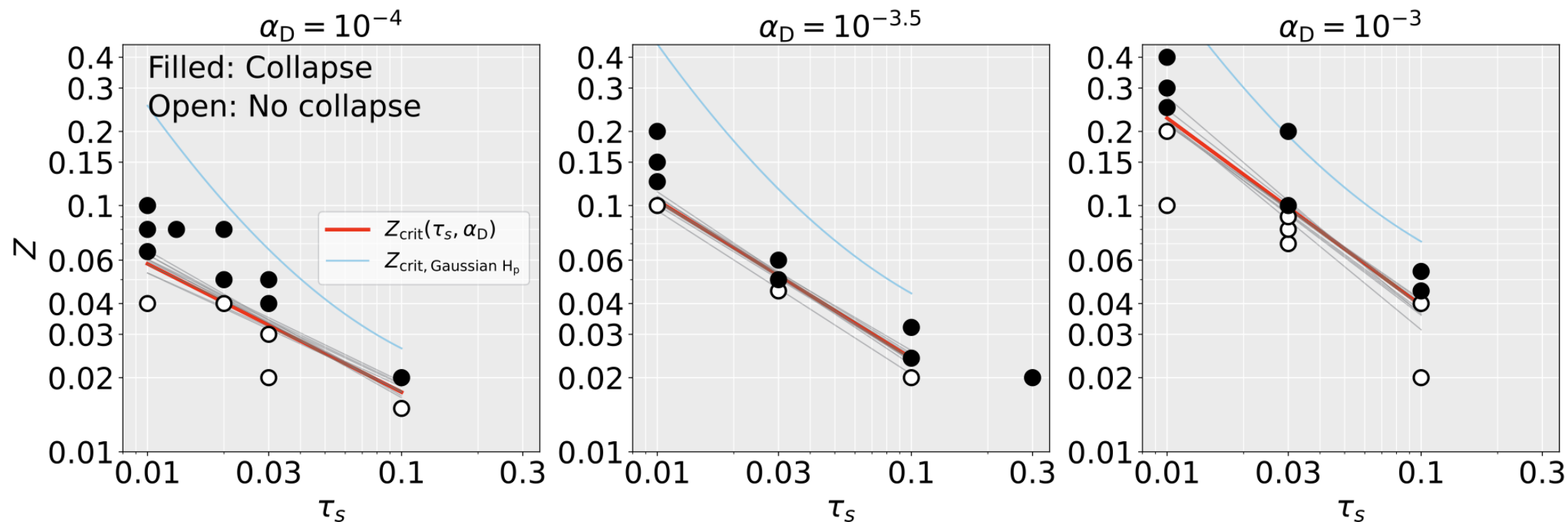
Main goal of simulations: find the parameter space conducive to strong clumping via the SI (find criteria for planetesimal formation)

- Youdin & Goodman (2005): linear stability analysis, $\epsilon \geq 1$ (linear, unstratified SI)
- Nonlinear SI, 2D: Carrera et al. (2015), Yang et al. (2017), Li & Youdin (2021), Lim et al. (2025a)
- Nonlinear SI, 3D: Lim et al. (2025b)



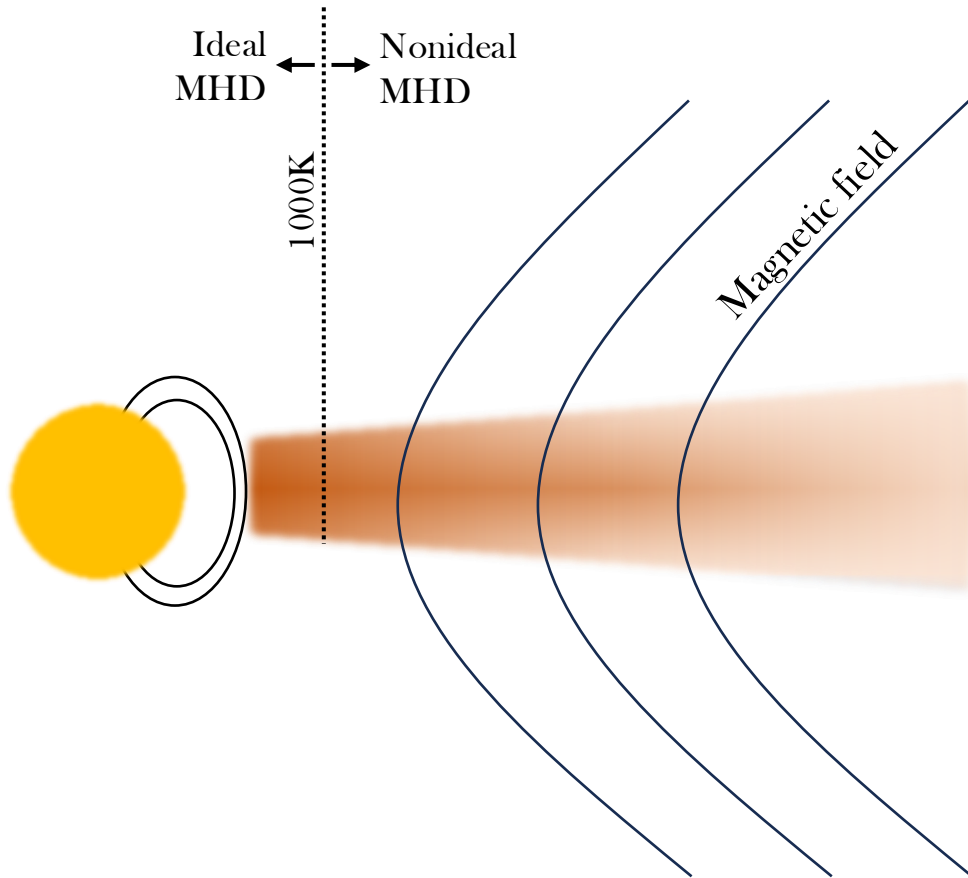
External turbulence

Lim et al. (2024): 3D stratified shearing box simulations with particle self-gravity and forced isotropic turbulence



External turbulence

Lim et al. (2024): 3D stratified shearing box simulations with particle self-gravity and forced isotropic turbulence



Eriksson et al. (2026): 3D stratified shearing box simulations with MRI-turbulence (non-ideal MHD with Ambipolar diffusion)

- Code: Athena (similar set-up as Xu & Bai 2022)
- 3D shearing box: $0.8H \times 0.8H \times 0.8H$
- Resolution: $250/H$
- Periodic boundary conditions
- Isothermal
- Vertically stratified dust, unstratified gas
- Pressure gradient: 0.05
- $Am = 1$
- Net vertical field: $\beta_0 = 6 \times 10^3, 6 \times 10^4$
- Lagrangian particles: one per cell
- No self-gravity

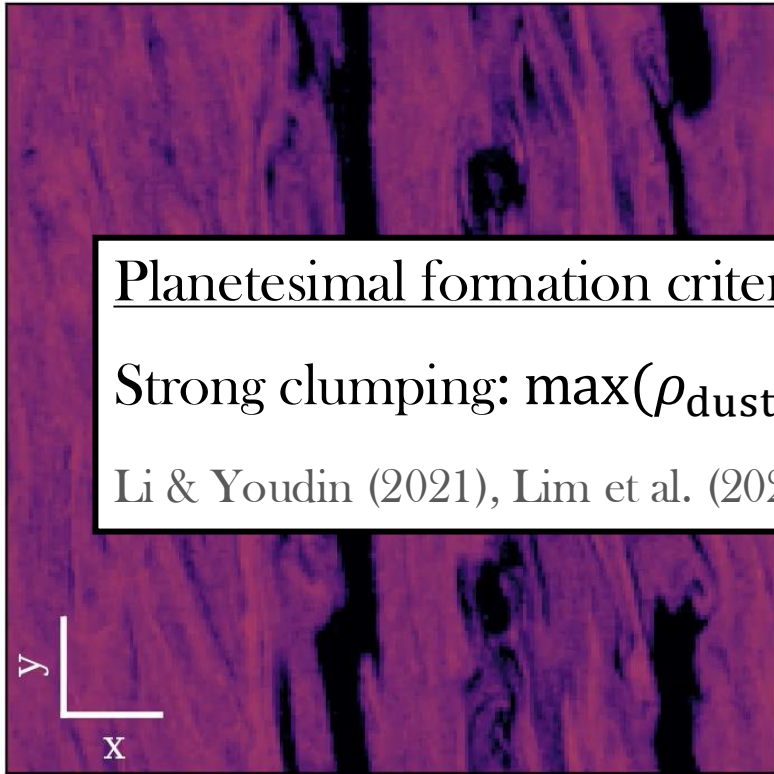
Face-on

Edge-on

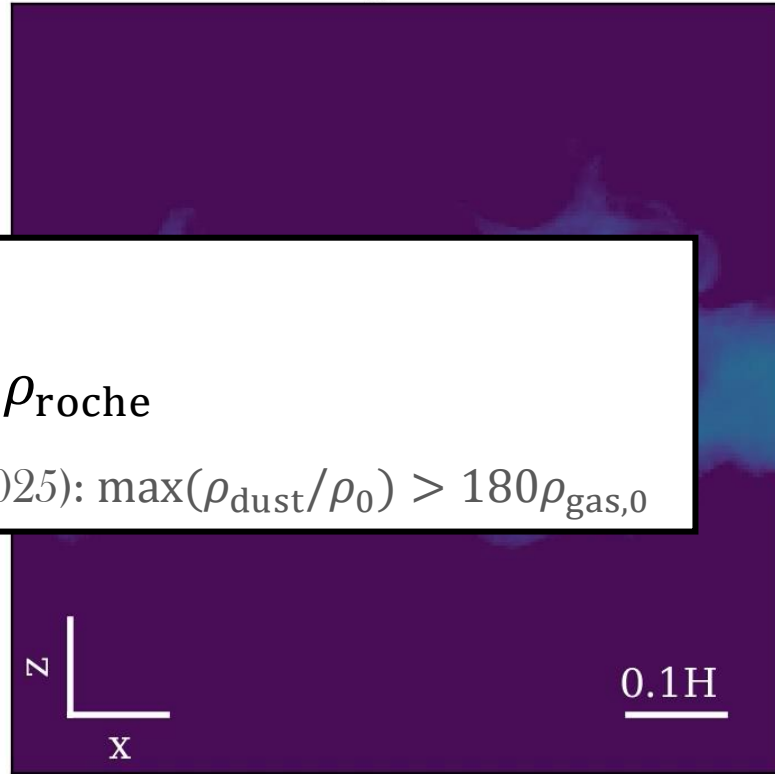
Planetesimal formation criteria

Strong clumping: $\max(\rho_{\text{dust}}) > \rho_{\text{roche}}$

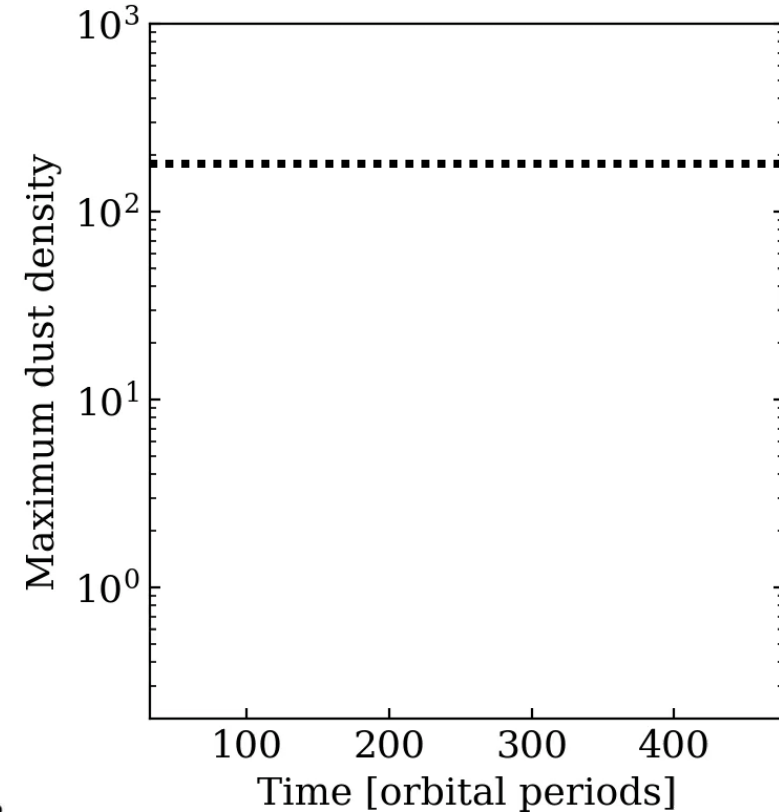
Li & Youdin (2021), Lim et al. (2024, 2025): $\max(\rho_{\text{dust}}/\rho_0) > 180\rho_{\text{gas},0}$



Midplane dust density



y-averaged dust density

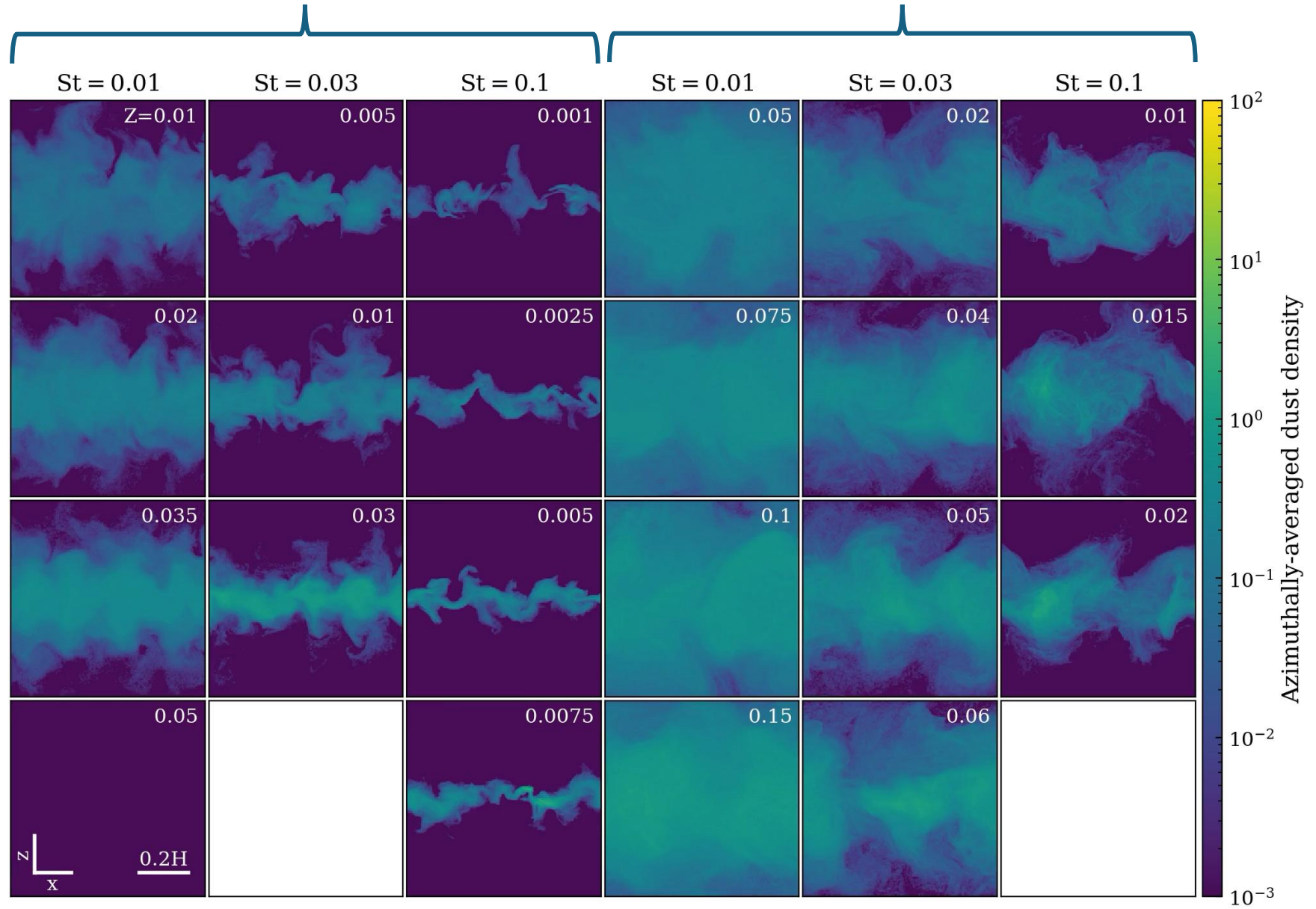


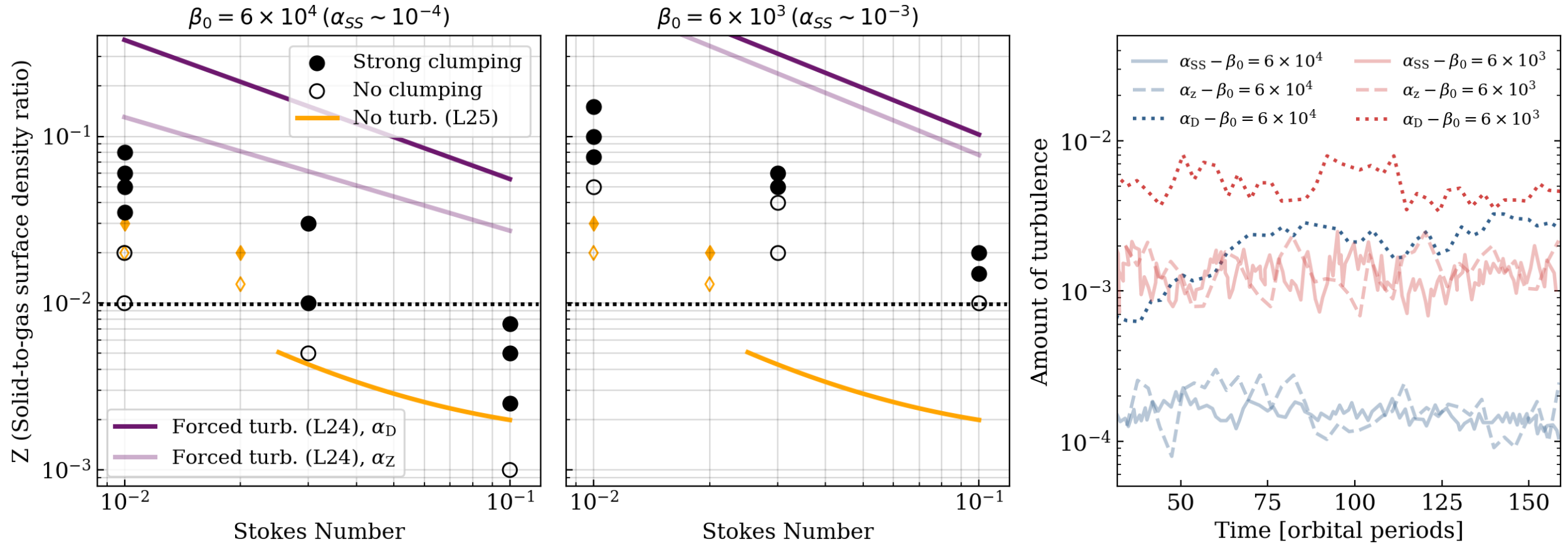
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$$\beta_0 = 6 \times 10^4 \ (\alpha_{SS} \sim 10^{-4})$$

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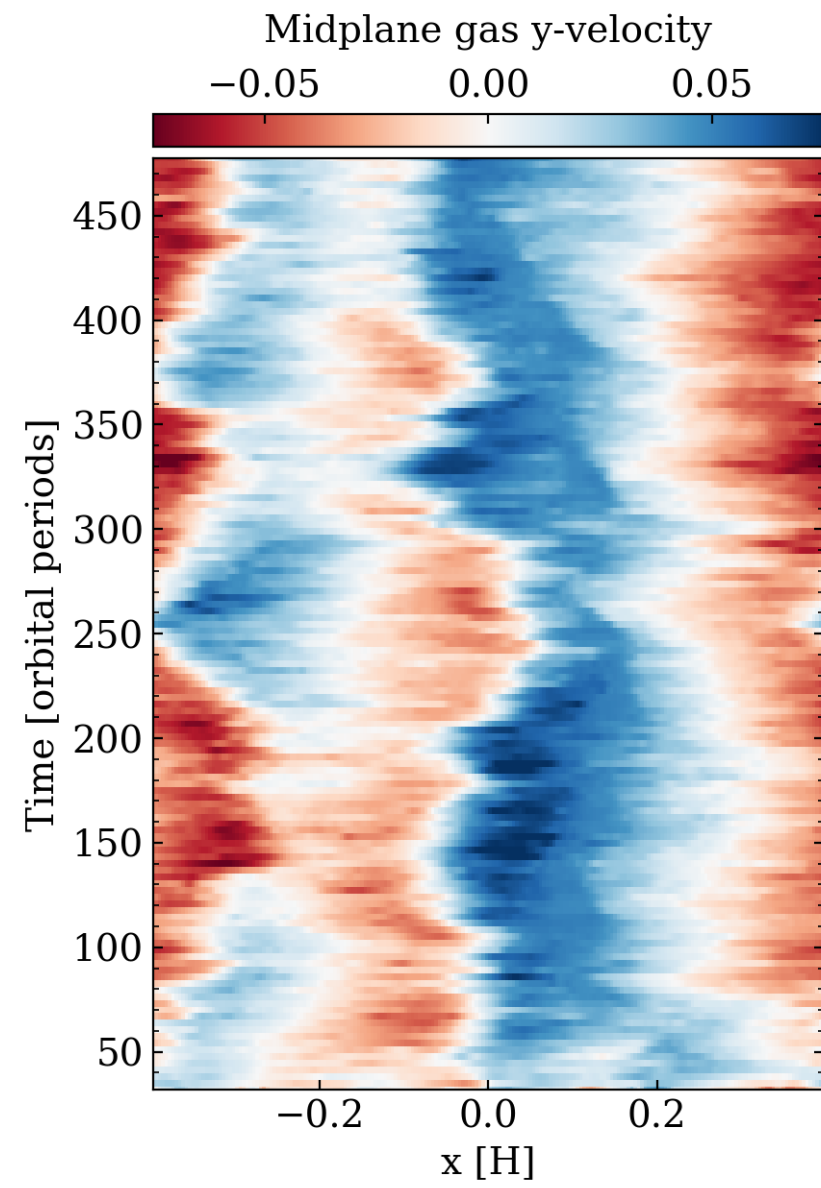
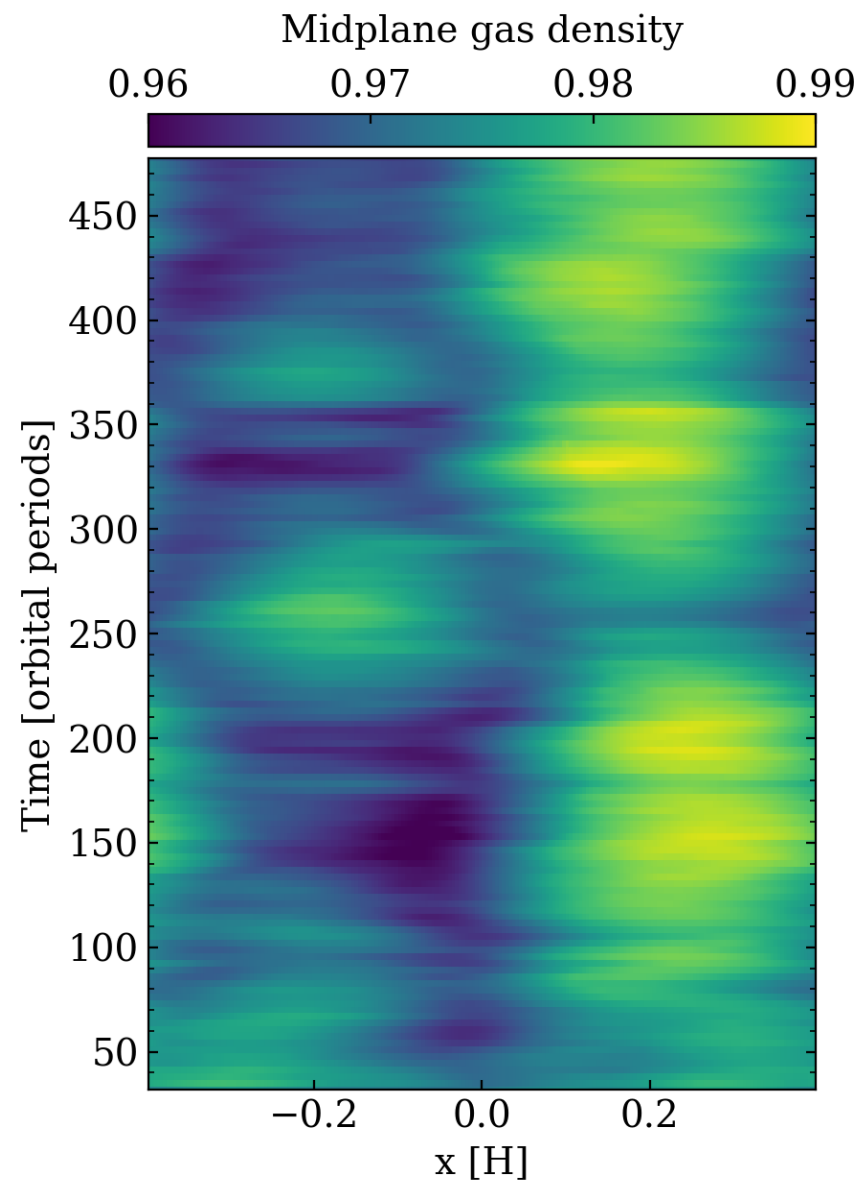
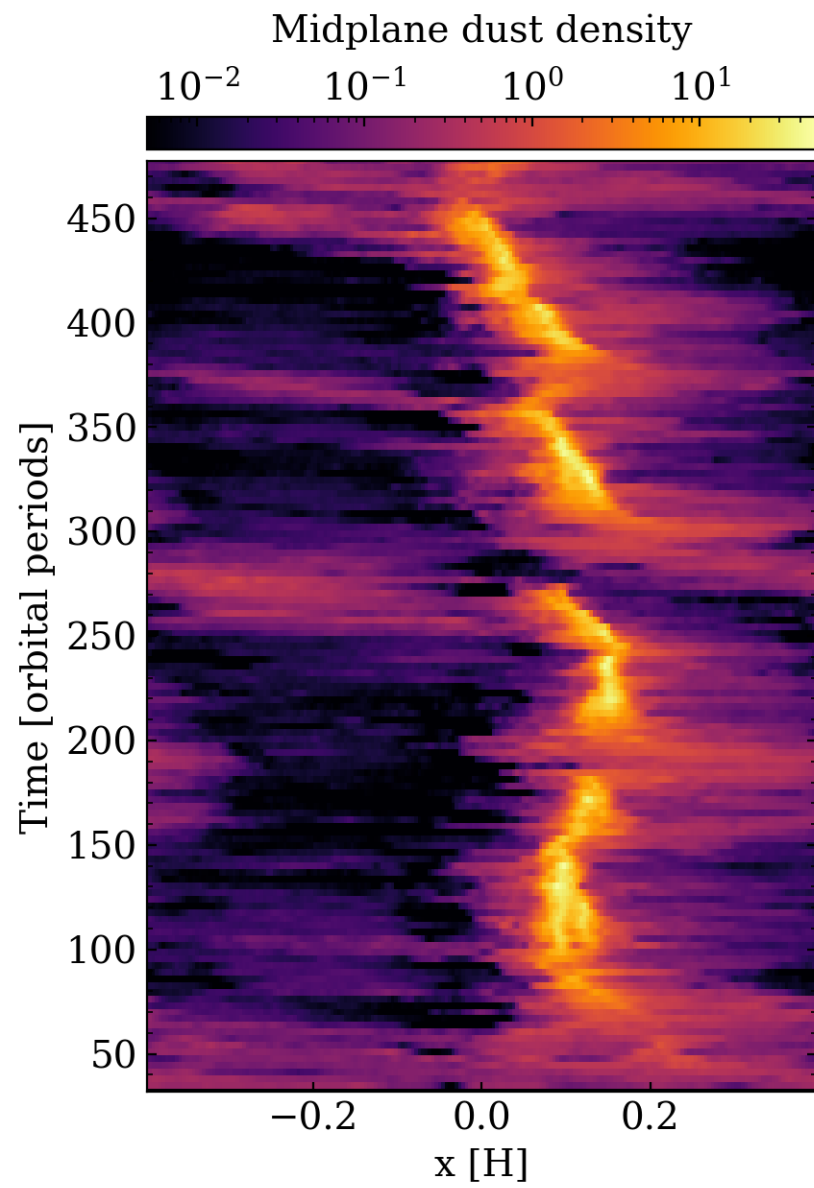
Higher dust-to-gas surface density ratio

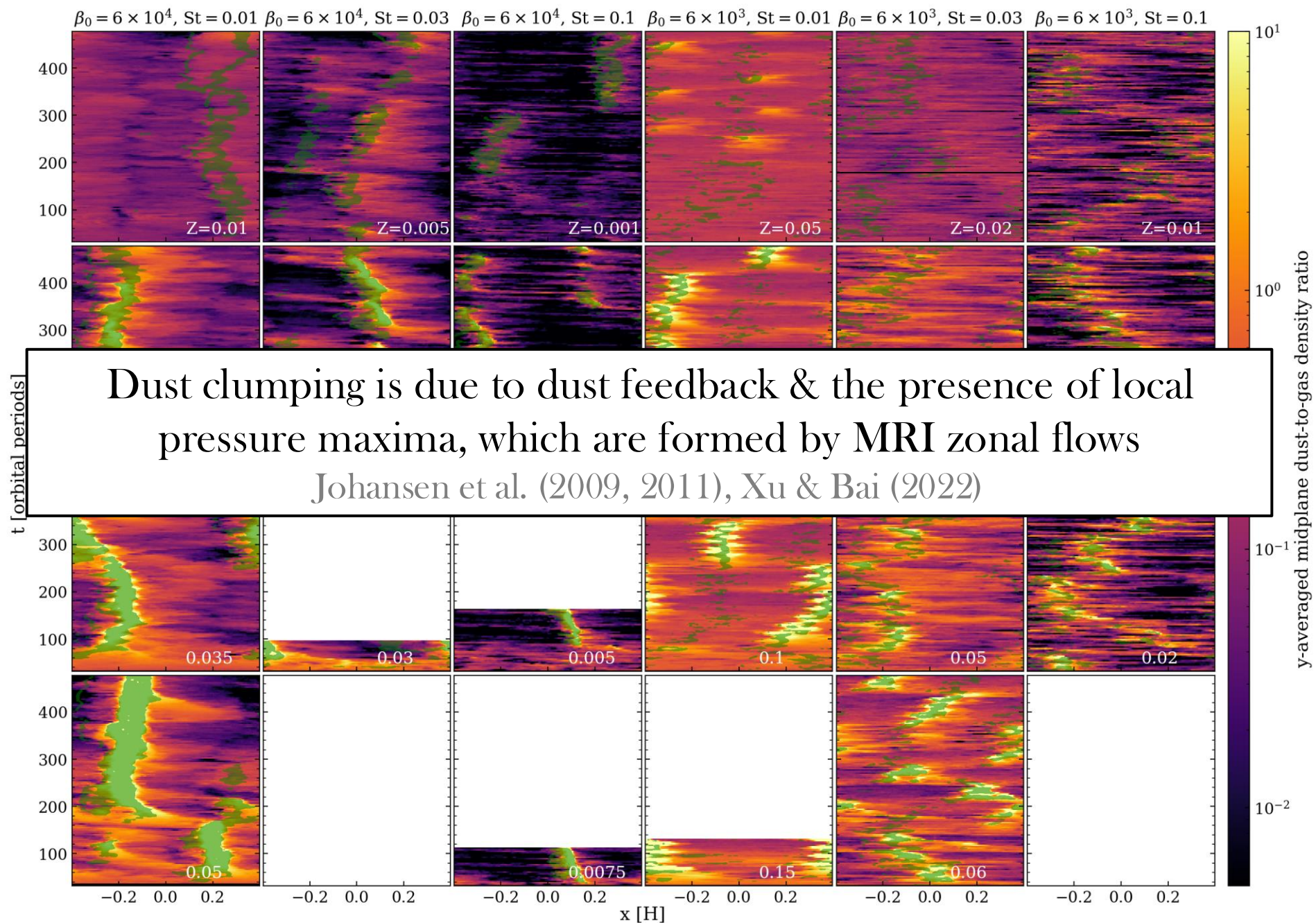




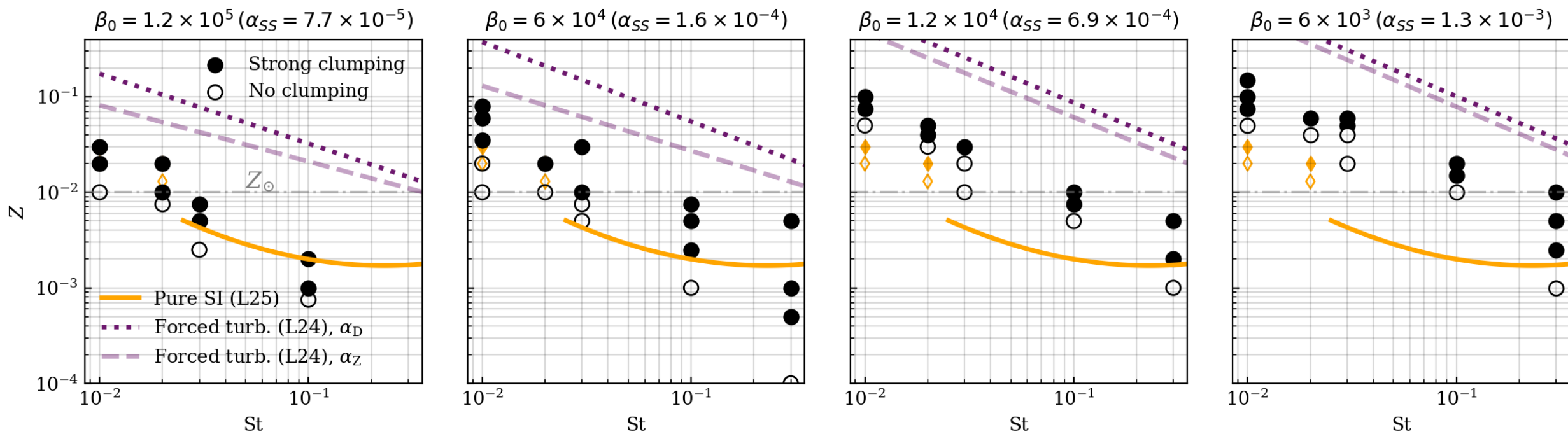
- Self-consistently driven MRI turbulence does not necessarily inhibit planetesimal formation via the SI
- Self-consistently driven MRI turbulence yield lower Z_{crit} than forced-turbulence models

Why?

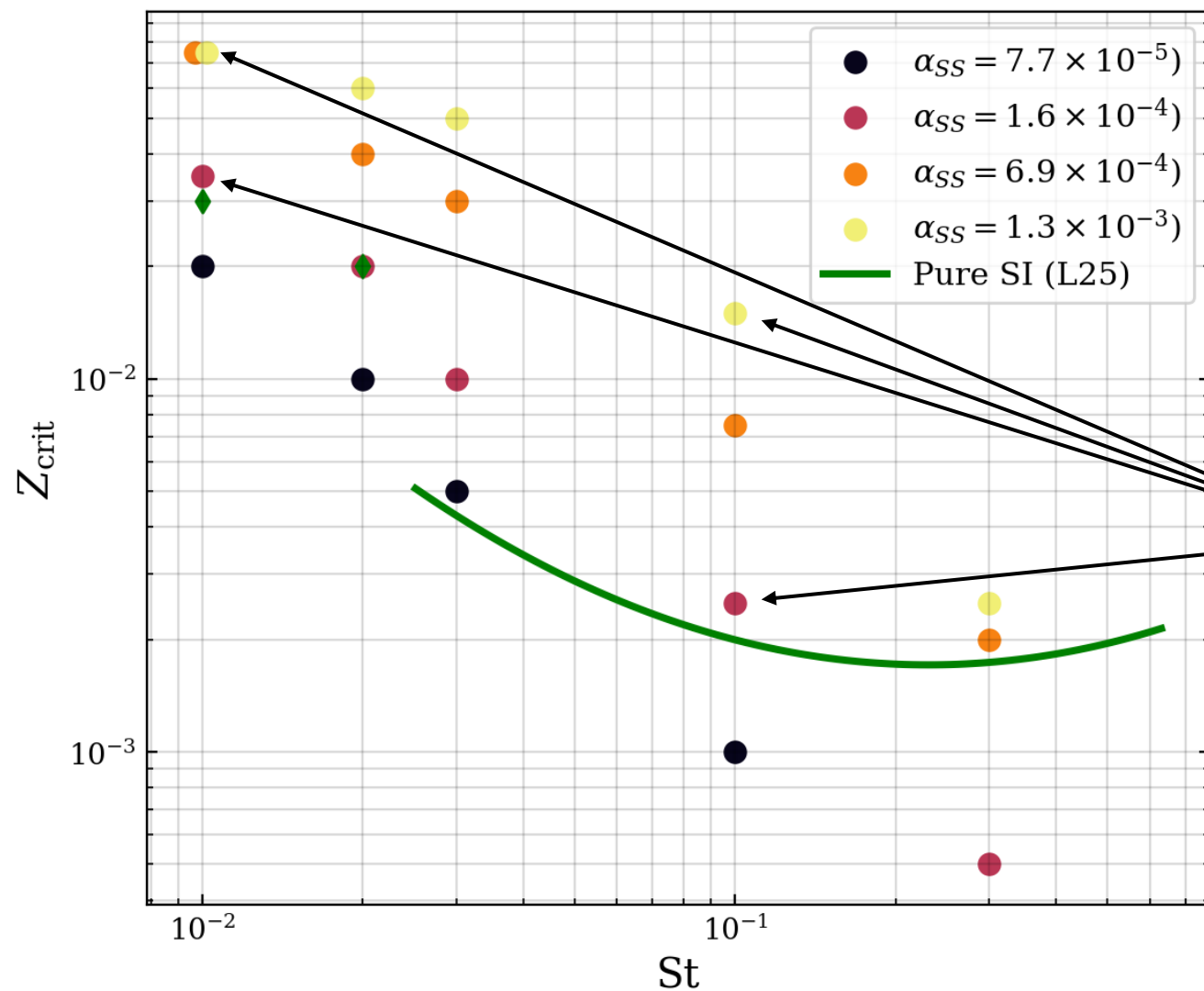




Work in progress



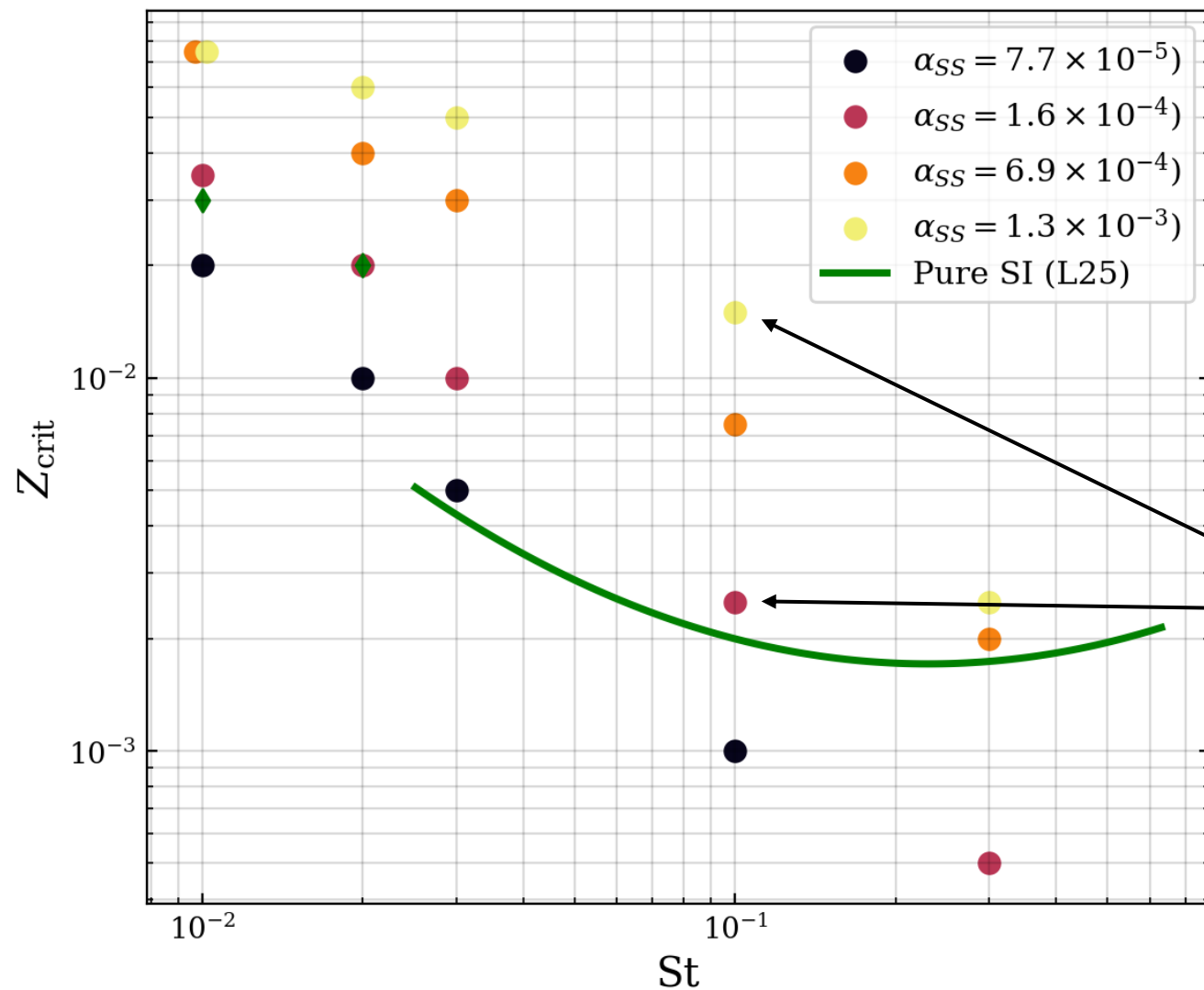
Work in progress



Convergence of Z_{crit} with resolution and box-size

- $L_x: 0.8H \rightarrow 1.4H$
- Resolution: $250/H \rightarrow 500/H$

Work in progress



Convergence of Z_{crit} with resolution and box-size

- $L_x: 0.8H \rightarrow 1.4H$
- Resolution: $250/H \rightarrow 500/H$

St = 0.01

St = 0.03

St = 0.1

St = 0.01

St = 0.03

St = 0.1

Z=0.01

0.005

0.001

0.05

0.02

0.01

Take-away points

- ✓ Self-consistently driven MRI turbulence does not necessarily inhibit planetesimal formation
- ✓ Self-consistently driven MRI turbulence yield lower Z_{crit} than forced-turbulence models
- ✓ Dust clumping is due to dust feedback & the presence of local pressure maxima, which are formed by MRI zonal flows

What's next?

- Fit for Z_{crit}
- Convergence of Z_{crit} with resolution
- Polydisperse SI – disks have a distribution of particle sizes